Matsubara Dynamics

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I

The Wigner transform of an operator A in position representation is

$$A_W = \int d\Delta \langle q - \frac{\Delta}{2}|A|q + \frac{\Delta}{2}\rangle e^{i\Delta p/\hbar} \tag{1.1}$$

one can show that it is equivalently in momentum representation as,

$$A_W = \int d\Lambda \langle p + \frac{\Lambda}{2}|A|p - \frac{\Lambda}{2}\rangle e^{iq\Lambda/\hbar} \tag{1.2}$$

one can futher show that

$$f(q)_W = f(q) \tag{1.3}$$

and

$$f(p)_W = f(p) \tag{1.4}$$

And Wigner transforms of Hermitian operators are real.
The trace of the product between two operators A and B can be written as

$$\mathrm{tr}[AB] = \frac{1}{2\pi\hbar}\int dp \int dq\, A_W B_W \tag{1.5}$$

The equation of motion of an observable in Heisenberg picture given by

$$\dot{A} = \mathcal{L}A \tag{1.6}$$

where we have defined the Liouvillian as

$$\mathcal{L}(\cdot) \equiv \frac{i}{\hbar}[H, \cdot] \tag{1.7}$$

One can also show that

$$\frac{i}{2m\hbar}[p^2, A]_W = \frac{p}{m}\frac{\partial}{\partial q}A_W \tag{1.8}$$

and

$$\frac{i}{\hbar}[V(q), A]_W = -\frac{2i}{\hbar}\int d\Delta \Big[\sinh\Big(\frac{\Delta}{2}\frac{\partial}{\partial q}\Big)V(q)\Big]\langle q - \frac{\Delta}{2}|A|q + \frac{\Delta}{2}\rangle e^{i\Delta p/\hbar} \tag{1.9}$$

If we proceed by the approximation $\sinh x \to x$, then we have

$$\frac{i}{\hbar}[V(q), A]_W = -\frac{\partial V}{\partial q}\frac{\partial}{\partial p}A_W \tag{1.10}$$

and the Liouvilian takes the classical form

$$\mathcal{L} = \frac{p}{m}\frac{\partial}{\partial q} - \frac{\partial V}{\partial q}\frac{\partial}{\partial p} \tag{1.11}$$

However, if we try to continue making process without the approximation, we obtain

$$\frac{i}{\hbar}[V(q), A]_W = -\frac{2}{\hbar}V(q)\sin\Big(\frac{\overleftarrow{\partial}}{\partial q}\frac{\hbar}{2}\frac{\overrightarrow{\partial}}{\partial p}\Big)A_W \tag{1.12}$$

Thus the full Liouvillian is

$$\mathcal{L} = \frac{p}{m}\frac{\partial}{\partial q} - \frac{2}{\hbar}V(q)\sin\Big(\frac{\overleftarrow{\partial}}{\partial q}\frac{\hbar}{2}\frac{\overrightarrow{\partial}}{\partial p}\Big) \tag{1.13}$$

Where the direction of the arrow tells us the direction which the differential operator acts on. Note that this is strictly speaking an abuse of notation, as we defined Liouvillian to act on operators rather directly on Wigner transforms, so what we really mean here is $(\mathcal{L}A)_W$ equals to RHS of Eq. (1.11) and (1.13) on A_W.

II

The Kubo-transformed real time correlation function is defined as

$$C_{AB}(t) \equiv \mathrm{tr}[A_\beta B(t)] \tag{2.1}$$

where

$$A_\beta \equiv \frac{1}{\beta}\int_0^\beta d\lambda\, e^{-(\beta-\lambda)H} A e^{-\lambda H} \tag{2.2}$$

Using Eq. (1.5) yields

$$C_{AB}(t) = \frac{1}{2\pi\hbar}\int dp \int dq\, A_{\beta W} B(t)_W \tag{2.3}$$

taking the time derivative yields

$$\dot{C}_{AB}(t) = \frac{1}{2\pi\hbar}\int dp \int dq\, A_{\beta W}\mathcal{L}[B(t)]_W \tag{2.4}$$

We now introduce the ring polymer real time correlation function as

$$C_{AB}^{[N]}(t) \equiv \iiint d\mathbf{q}\, d\mathbf{\Delta}\, d\mathbf{z}\, A(\mathbf{q})B(\mathbf{z})\prod_{l=1}^{N}\langle q_{l-1}-\frac{\Delta_{l-1}}{2}|e^{-\beta_N H}|q_l+\frac{\Delta_l}{2}\rangle\langle q_l+\frac{\Delta_l}{2}|e^{-iHt/\hbar}|z_l\rangle\langle z_l|e^{iHt/\hbar}|q_l-\frac{\Delta_l}{2}\rangle \tag{2.5}$$

where $\beta_N \equiv \beta/N$ and

$$A(\mathbf{q}) \equiv \frac{1}{N}\sum_{i=1}^{N} A(q_i) \tag{2.6}$$

and similarly for $B(\mathbf{q})$ and when these are functions of $\mathbf{p}$.
One can show that

$$C_{AB}(t) = \lim_{N\to\infty} C_{AB}^{[N]}(t) \tag{2.7}$$

Furthermore, the correlation function can be rewritten as

$$C_{AB}^{[N]}(t) = \frac{1}{(2\pi\hbar)^N}\int d\mathbf{p}\int d\mathbf{q}\left[e^{-\beta H}A\right]_{\bar{N}}(\mathbf{p},\mathbf{q})B(t)_N(\mathbf{p},\mathbf{q}) \tag{2.8}$$

where the subscripts N and $\bar{N}$ denotes generalised Wigner transforms

$$\left[e^{-\beta H}A\right]_{\bar{N}}(\mathbf{p},\mathbf{q}) \equiv \int d\mathbf{\Delta}\, A(\mathbf{q})\prod_{l=1}^{N}\langle q_{l-1} - \frac{\Delta_{l-1}}{2}|e^{-\beta_N H}|q_l + \frac{\Delta_l}{2}\rangle e^{ip_l\Delta_l/\hbar} \tag{2.9}$$

$$B(t)_N(\mathbf{p},\mathbf{q}) \equiv \int d\mathbf{\Delta}\int d\mathbf{z}\, B(\mathbf{z})\prod_{l=1}^{N}\langle q_{l-1} - \frac{\Delta_{l-1}}{2}|e^{-iHt/\hbar}|z_l\rangle\langle z_l|e^{iHt/\hbar}|q_l + \frac{\Delta_l}{2}\rangle e^{ip_l\Delta_l/\hbar} \tag{2.10}$$

The time derivative is

$$\dot{C}_{AB}^{[N]}(t) = \frac{1}{(2\pi\hbar)^N}\int d\mathbf{p}\int d\mathbf{q}\left[e^{-\beta H}A\right]_{\bar{N}} L_N[B(t)]_N \tag{2.11}$$

where

$$L_N \equiv \sum_{l=1}^{N}\frac{p_l}{m}\frac{\partial}{\partial q_l} - \frac{2}{\hbar}V(q_l)\sin\Big(\frac{\overleftarrow{\partial}}{\partial q_l}\frac{\hbar}{2}\frac{\overrightarrow{\partial}}{\partial p_l}\Big) \tag{2.12}$$

which is a generalisation of Eq. (1.13), and can be written more compactly as

$$L_N = \frac{1}{m}\mathbf{p}\cdot\nabla_{\mathbf{q}} - \frac{2}{\hbar}U_N(\mathbf{q})\sin\Big(\frac{\hbar}{2}\overleftarrow{\nabla_{\mathbf{q}}}\cdot\overrightarrow{\nabla_{\mathbf{p}}}\Big) \tag{2.13}$$

where $U_N(\mathbf{q})$ in Eq. (2.6).

III

We now introduce the normal mode coordinate for the ring polymer

$$Q_n = \sum_{l=1}^{N} T_{ln} q_l \tag{3.1}$$

we let n be odd for the sake of simplicity and $n = 0, \pm1, \ldots, \pm(N-1)/2$ and similarly for P_n.

T_{ln} is an orthogonal matrix

$$T_{ln} = \begin{cases} N^{-1/2} & n = 0 \\ \sqrt{2/N}\sin(2\pi ln/N) & n = 1, \ldots, (N-1)/2 \\ \sqrt{2/N}\cos(2\pi ln/N) & n = -1, \ldots, -(N-1)/2 \end{cases} \tag{3.2}$$

with normal mode frequencies

$$\omega_n = \frac{2}{\beta_N \hbar}\sin\Big(\frac{n\pi}{N}\Big) \tag{3.3}$$

Eq. (2.11) can now be written in normal mode coordinate

$$\dot{C}^{[N]}_{AB}(t) = \frac{1}{(2\pi\hbar)^N}\int d\mathbf{P}\int d\mathbf{Q}\left[e^{-\beta H}A\right]_{\tilde{N}} L_N B(t)_N \tag{3.4}$$

and also the Liouvillian

$$L_N = \frac{1}{m}\mathbf{P}\cdot\nabla_{\mathbf{Q}} - \frac{2}{\hbar}U_N(\mathbf{Q})\sin\Big(\frac{\hbar}{2}\overleftarrow{\nabla_{\mathbf{Q}}}\cdot\overrightarrow{\nabla_{\mathbf{P}}}\Big) \tag{3.5}$$

Consider the limit of the ring polymer with large number of beads N, such that the lowest modes $|n| \leq (M-1)/2$ and $M \ll N$, and we have

$$\tilde{\omega}_n \equiv \lim_{N\to\infty}\omega_n = \frac{2n\pi}{\beta\hbar} \tag{3.6}$$

Matsubara modes have the special property that any superposition of them produces a distribution of q_l such that it is a smooth path in imaginary time. Mixing in non-Matsubara modes would result in a jagged path. From this point all quantities with ˜ on top of its head are associated with Matsubara modes.

$$\tilde{Q}_n \equiv \lim_{N\to\infty}\frac{Q_n}{\sqrt{N}} \tag{3.7}$$

Now we split the Liouvillian into two parts

$$\lim_{N\to\infty} L_N = \mathcal{L}_M + \lim_{N\to\infty} L_{\text{err}}(N, M) \tag{3.8}$$

where the Matsubara Liouvillian is defined as

$$\mathcal{L}_M \equiv \lim_{N\to\infty}\frac{1}{m}\mathbf{P}\cdot\nabla_{\mathbf{Q}} - \frac{2}{\hbar}U_N(\mathbf{Q})\sin\Big(\frac{\hbar}{2}\overleftarrow{\nabla_{\mathbf{Q}}}\cdot\overrightarrow{\nabla_{\mathbf{P}}}\Big) \tag{3.9}$$

Where all the vector quantities now have dimension M equals to the number of lowest frequency modes rather than the full dimension N like in Eq. (3.5).

We can make significant simplification by using the fact that $\tilde{U}_M(\tilde{\mathbf{Q}}) = U_N(\mathbf{Q})/N$

$$\mathcal{L}_M = \lim_{N\to\infty}\frac{1}{m}\tilde{\mathbf{P}}\cdot\nabla_{\tilde{\mathbf{Q}}} - \frac{2N}{\hbar}\tilde{U}_M(\tilde{\mathbf{Q}})\sin\Big(\frac{\hbar}{2N}\overleftarrow{\nabla_{\tilde{\mathbf{Q}}}}\cdot\overrightarrow{\nabla_{\tilde{\mathbf{P}}}}\Big) \tag{3.10}$$

Taylor series expansion of sin yields

$$\mathcal{L}_M = \lim_{N\to\infty}\frac{1}{m}\tilde{\mathbf{P}}\cdot\nabla_{\tilde{\mathbf{Q}}} - \tilde{U}_M(\tilde{\mathbf{Q}})\overleftarrow{\nabla}_{\tilde{\mathbf{Q}}}\cdot\overrightarrow{\nabla}_{\tilde{\mathbf{P}}} \tag{3.11}$$

The error Liouvillian is given by the higher frequency non-Matsubara modes

$$L_{\rm err}(N,M) \equiv \sum_{n=(M+1)/2}^{(N-1)/2} \frac{P_{-n}}{m}\frac{\partial}{\partial Q_{-n}} + \frac{P_n}{m}\frac{\partial}{\partial Q_n} - \frac{4}{\hbar}U(\mathbf{Q})\sin\Big(\frac{X}{2}\Big)\cos\Big(Y+\frac{X}{2}\Big) \tag{3.12}$$

where

$$X \equiv \frac{\hbar}{2}\sum_{n=(M+1)/2}^{(N-1)/2} \frac{\overleftarrow{\partial}}{\partial Q_{-n}}\frac{\overrightarrow{\partial}}{\partial P_{-n}} + \frac{\overleftarrow{\partial}}{\partial Q_n}\frac{\overrightarrow{\partial}}{\partial P_n} \tag{3.13}$$

$$Y \equiv \frac{\hbar}{2}\sum_{n=(M+1)/2}^{(N-1)/2} \frac{\overleftarrow{\partial}}{\partial Q_n}\frac{\overrightarrow{\partial}}{\partial P_n} \tag{3.14}$$

The Liouvillian is exact until this point would generate the exact quantum dynamics, however, as one can see that it is an extremely complicated object and one must make an approximation to hope that it can be applied in any practical setting. We choose to drop out the $L_{\rm err}$ term as our first approximation, this leads to what is known as the **Matsubara dynamics**. The dynamics ensures the distribution of $(\mathbf{p},\mathbf{q})$ is a smooth and diffrentiable function of imaginary time τ at $t=0$ and $t>0$.

The Kubo-transformed real time correlation function is therefore

$$C_{AB}^{[M]}(t) = \lim_{N\to\infty}\frac{1}{(2\pi\hbar)^N}\int d\mathbf{P}\int d\mathbf{Q}\left[e^{-\beta H}A\right]_{\bar{N}} e^{\mathcal{L}_M t}[B(0)]_N \tag{3.15}$$

and the exact Matsubara correlation function is the limit

$$C_{AB}(t) \simeq C_{AB}^{\rm Mats}(t) = \lim_{M\to\infty} C_{AB}^{[M]}(t) \tag{3.16}$$

under the contraint $M \ll N$.

Which can be written as

$$C_{AB}^{[M]}(t) = \frac{\alpha_M}{2\pi\hbar}\int d\tilde{\mathbf{P}}\int d\tilde{\mathbf{Q}}\, A(\tilde{\mathbf{Q}})e^{-\beta[\tilde{H}_M(\tilde{\mathbf{P}},\tilde{\mathbf{Q}})-i\theta_M(\tilde{\mathbf{P}},\tilde{\mathbf{Q}})]}e^{\mathcal{L}_M t}B(\tilde{\mathbf{Q}}) \tag{3.17}$$

where $\alpha_M = \hbar^{1-M}[(M-1)/2]!^2$ and the Matsubara Hamiltonian is

$$\tilde{H}_M(\tilde{\mathbf{P}},\tilde{\mathbf{Q}}) = \frac{\tilde{\mathbf{P}}^2}{2m} + \tilde{U}_M(\tilde{\mathbf{Q}}) \tag{3.18}$$

and the phase is

$$\tilde{\theta}_M(\tilde{\mathbf{P}},\tilde{\mathbf{Q}}) = \sum_{n=-(M-1)/2}^{(M-1)/2} \tilde{P}_n\tilde{\omega}_n\tilde{Q}_{-n} \tag{3.19}$$

Upon the coordinate transformation

$$\bar{P}_n = \tilde{P}_n - im\tilde{\omega}_n\tilde{Q}_{-n} \tag{3.20}$$

we have

$$C_{AB}^{[M]}(t) = \frac{\alpha_M}{2\pi\hbar}\left[\prod_{n=-(M-1)/2}^{(M-1)/2}\int_{-\infty-im\tilde{\omega}_n\tilde{Q}_{-n}}^{\infty-im\tilde{\omega}_n\tilde{Q}_{-n}} d\bar{P}_n\right]\int d\tilde{\mathbf{Q}}\, A(\tilde{\mathbf{Q}})e^{-\beta\tilde{R}_M(\bar{\mathbf{P}},\tilde{\mathbf{Q}})]}e^{\mathcal{L}_M t}B(\tilde{\mathbf{Q}}) \quad (3.21)$$

where the ring polymer Hamiltonian is

$$\tilde{R}_M(\bar{\mathbf{P}},\tilde{\mathbf{Q}}) = \sum_{n=-(M-1)/2}^{(M-1)/2}\left(\frac{\bar{P}_n^2}{2m} + \frac{1}{2}m\tilde{\omega}_n^2\tilde{Q}_n^2\right) + \tilde{U}_M(\tilde{\mathbf{Q}}) \quad (3.22)$$

At $t = 0$, as integrand is analytic and and vanishes when real part of $\bar{P}_n$ is large, then we can shift the integral onto the real axis using Cauchy integral theorem

$$C_{AB}^{[M]}(0) = \frac{\alpha_M}{2\pi\hbar}\int d\bar{\mathbf{P}}\int d\tilde{\mathbf{Q}}\, A(\tilde{\mathbf{Q}})e^{-\beta\tilde{R}_M(\bar{\mathbf{P}},\tilde{\mathbf{Q}})]}B(\tilde{\mathbf{Q}}) \quad (3.23)$$

However, the contour integration trick does not work for $t > 0$.

IV

Using the change of variable Eq. (3.18), one can rewrite the Matsubara Liouvillian as

$$\mathcal{L}_M = \mathcal{L}_M^{[RP]} + i\mathcal{L}^{[I]} \quad (4.1)$$

where

$$\mathcal{L}_M^{[RP]} \equiv \sum_{n=-(M-1)/2}^{(M-1)/2}\frac{\bar{P}_n}{2m}\frac{\partial}{\partial\tilde{Q}_n} - \left[m\tilde{\omega}_n^2\tilde{Q}_n + \frac{\partial\tilde{U}_M(\tilde{\mathbf{Q}})}{\partial\tilde{Q}_n}\right]\frac{\partial}{\partial\bar{P}_n} \quad (4.2)$$

and

$$\mathcal{L}_M^{[\mathrm{I}]} \equiv \sum_{n=-(M-1)/2}^{(M-1)/2}\tilde{\omega}_n\left(\bar{P}_n\frac{\partial}{\partial\bar{P}_{-n}} - \tilde{Q}_n\frac{\partial}{\partial\tilde{Q}_{-n}}\right) \quad (4.3)$$

It is possible to follow a path along which one gradually moves $\bar{P}_n$, $n \neq 0$, towards the real axis whilst gradually discarding $\mathcal{L}_M^{[\mathrm{I}]}$, such that the dynamics remains stable (and the contour-integration trick remains valid) at every point along the path. At the end of the path, $\mathcal{L}_M^{[\mathrm{I}]}$ has been completely discarded, and $\bar{P}_n$ has reached the real axis. This results in the approximation,

$$C_{AB}^{[M]}(t) \simeq \frac{\alpha_M}{2\pi\hbar}\int d\bar{\mathbf{P}}\int d\tilde{\mathbf{Q}}\, A(\tilde{\mathbf{Q}})e^{-\beta\tilde{R}_M(\bar{\mathbf{P}},\tilde{\mathbf{Q}})}e^{\mathcal{L}_M^{[\mathrm{RP}]}t}B(\tilde{\mathbf{Q}}) \quad (4.4)$$

which is **ring polymer molecular dynamics**.